

PANDA Component Description: *ext_interrupt*

Overview

The *ext_interrupt* component implements an external interrupt which can be triggered on one or more of the following events: rising edge, falling edge, high state or low state at the interrupt pin.

The component can be configured to synchronize the external interrupt signal or not over a generic.

Generics

Name	Type	Default	Description
synchronize	boolean	true	true → synchronize external interrupt signal false → don't synchronize external interrupt signal

External Ports

Name	Type	Width	Description
ext_int	std_logic	1	External interrupt signal

Registers

Trigger Register

Name: TRIG

Address: Base + 0x00

Name	PIN	-	-	-	LOW	HIGH	FALL	RISE
R/W	R	-	-	-	R/W	R/W	R/W	R/W
Reset Value	-	0	0	0	0	0	0	0

The *TRIG* register shows all events which were triggered since their last clear and the state of the external interrupt signal. Each triggered event can be cleared by writing a '0' to the associated bit. Writing a '1' to an event bit does not change its state. Therefore all event bits can only be cleared but not set by software.

If a bit in *TRIG* is set and the corresponding bit in *ENA* is set too, an interrupt is generated. If all bits in *TRIG* are cleared which are set in *ENA*, then the interrupt is cleared.

RISE:	'1'	→	A rising edge occurred on <i>ext_int</i>
	'0'	→	No rising edge occurred on <i>ext_int</i> since <i>RISE</i> was cleared
FALL:	'1'	→	A falling edge occurred on <i>ext_int</i>
	'0'	→	No falling edge occurred on <i>ext_int</i> since <i>FALL</i> was cleared
HIGH:	'1'	→	A high state occurred on <i>ext_int</i>
	'0'	→	No high state occurred on <i>ext_int</i> since <i>HIGH</i> was cleared
LOW:	'1'	→	A low state occurred on <i>ext_int</i>
	'0'	→	No low state occurred on <i>ext_int</i> since <i>LOW</i> was cleared
PIN:	Current state of the <i>ext_int</i> pin.		

Enable Register

Name: ENA

Address: Base + 0x01

Name	-	-	-	-	LOW	HIGH	FALL	RISE
R/W	-	-	-	-	R/W	R/W	R/W	R/W
Reset Value	0	0	0	0	0	0	0	0

The *ENA* register defines which events can trigger an interrupt. If a bit is being set while the corresponding bit in *TRIG* was already set before, an interrupt is generated immediately. If all Bits in *ENA* are cleared which are set in *TRIG*, the interrupt is cleared.

RISE: '1' → An interrupt is generated if *RISE* in *TRIG* is set
'0' → No interrupt is generated if *RISE* in *TRIG* is set

FALL: '1' → An interrupt is generated if *FALL* in *TRIG* is set
'0' → No interrupt is generated if *FALL* in *TRIG* is set

HIGH: '1' → An interrupt is generated if *HIGH* in *TRIG* is set
'0' → No interrupt is generated if *HIGH* in *TRIG* is set

LOW: '1' → An interrupt is generated if *LOW* in *TRIG* is set
'0' → No interrupt is generated if *LOW* in *TRIG* is set